

Seattle and Baltimore Rain Comparison

I. Introduction

This report aims to determine if it rains more in Seattle than Baltimore, MD. There are a couple of different approaches to determine if it rains more in one city versus another. ‘More’ can refer to either the measurement of rainfall or the frequency of days when it rains. However, conceptualizing a given numerical amount of rainfall can be difficult, which makes the measurement of rainfall slightly less applicable. However, the amount of rainfall can give some indication to the duration of rain. If we assume, in general, the larger the amount of rainfall, the longer the rain event could have occurred. Therefore, it is also important to take into account the number of rainy days or proportion of days with rain. This report aims to compare the rainfall in Baltimore, MD and Seattle, WA by comparing the amount of rainfall and proportion of days with rain over a 5-year period.

II. Analysis & Results

This analysis uses data from NOAA National Centers for Environmental Information. The Climate Data Online provides access to an archive of historical weather and climate data for a specified weather station. The selected timeframe for this analysis is January 1st, 2018, to December 31st, 2022. NOAA provides several various weather-related observations; however, this analysis will focus on precipitation, measured in inches. The selected data frames contained observations for Seattle, WA and Baltimore, MD. It was confirmed that the data included only one weather station per city. Columns were re-formatted for consistency in values and clarity of titles. To address null date values, a list of all the dates in the target period was generated and added to the Baltimore data frame, setting null values for precipitation. The Baltimore and Seattle data sets were then merged, including only the date and precipitation columns, and reformatted to a tidy data frame.

Then, an algorithm was created to approximate missing precipitation values. This was done by assigning every date a ‘day of year’ number (January 1st = 1, January 2nd = 2, and so on). Each null precipitation observation was replaced with the average amount of precipitation for its corresponding city and day of year. There were two other additional variables created, month and any precipitation. The consequent analysis focused on answering the following questions: Which city has a higher amount of precipitation? Which city has a higher proportion of rainy days?

To compare the amount of precipitation, mean and 5-number summary were calculated for the two cities. Seattle has an average daily precipitation of 0.113 inches with a standard deviation of 0.241 inches. Baltimore has an average daily precipitation of 0.142 inches with a standard deviation of 0.377 inches. Seattle has an overall lower average precipitation. Baltimore has a considerably higher maximum precipitation value; 4.79 inches compared to 2.60 inches. Figure 1 shows the difference in average daily precipitation values between the two cities.

The mean monthly precipitation was then calculated by averaging every observation within each month across all 5 years. A t-test was performed to determine the statistical significance of the difference in average monthly precipitation. Figure 2 shows the average precipitation per month for each city. The statistically significant differences are indicated with an asterix. Seattle appears to

have a higher precipitation amount in the winter months, comparable amounts of precipitation in the spring and fall, lower amounts of precipitation in the summer. However, the months with a statistically significant difference are January, July, August, and September.

To compare the frequency of rainy days, the ratio of days with precipitation was calculated. Overall, Seattle has a higher proportion of days with precipitation than Baltimore. Figure 3 shows the proportion of days with precipitation for each city. A z-test was performed to determine the statistical significance of the difference in proportion of days with rain for the two cities by month. The majority of months had a statistically significant difference except for March, June, and September. Figure 2 shows the proportion of rainy days per month for the two cities, with an asterisk for statistically significant differences.

III. Discussion

Interestingly, this analysis has demonstrated a complicated conclusion. It is difficult to definitively say which city rains more. If we compare overall average rainfall, the answer is Baltimore; if we compare the proportion of days with rainfall, the answer is Seattle. In terms of average amount of rainfall, Baltimore has a higher average daily amount of rainfall and a higher maximum rainfall. Baltimore often tends to have a more consistent monthly average rainfall than Seattle. In Figure 3, Baltimore's average daily precipitation hovers around 0.15 per month. Whereas Seattle exhibits more severe changes in average precipitation depending on the time of year. Additionally, Seattle also exhibits a much wider range of monthly proportions of rainy days. In Figure 4, Seattle's proportion of days with rainfall fluctuates between approximately 0.2 to 0.8, depending on the time of year. When it rains in Baltimore, it tends to result in a higher amount of rainfall than in Seattle. Seattle appears to be more prone to frequent but lower amounts of rainfall and exhibits more seasonal changes in precipitation.

IV. Limitations

This report was limited by the lack of timestamps for weather observations. Therefore, it is difficult to estimate the intensity and duration of rainfall events. For example, a daily total of 1 inch of rainfall over a longer period may indicate a light rain event, despite the duration being longer. 1 inch of rainfall over 2 hours would indicate a much more intense rain event and be more likely to have a bigger impact on the city. Understanding these additional variables would result in a more extensive understanding of the rain in each city.

V. Appendix

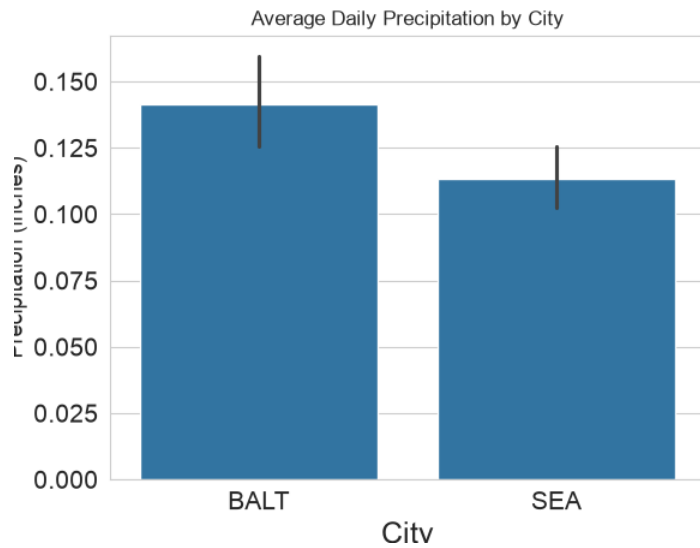


Figure 1: ‘Average Daily Precipitation by City.’ Shows the average daily precipitation amounts in inches by city, including the confidence intervals. Baltimore is on the left and Seattle is on the right.

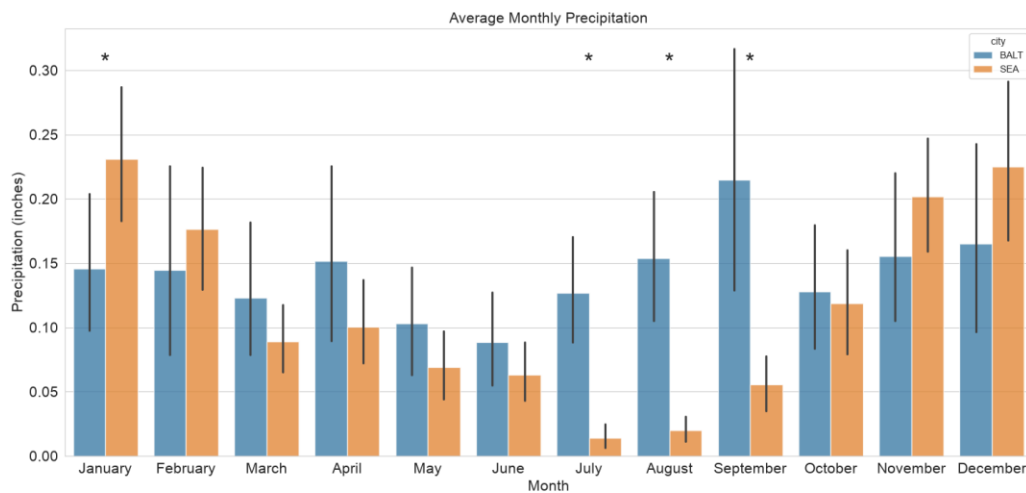


Figure 3: “Average Monthly Precipitation.” This graph shows the average precipitation across 5 years by month and city. It includes confidence intervals and asterixis above months determined to have a statistically significant difference in average precipitation between the Baltimore, blue, and Seattle, orange.

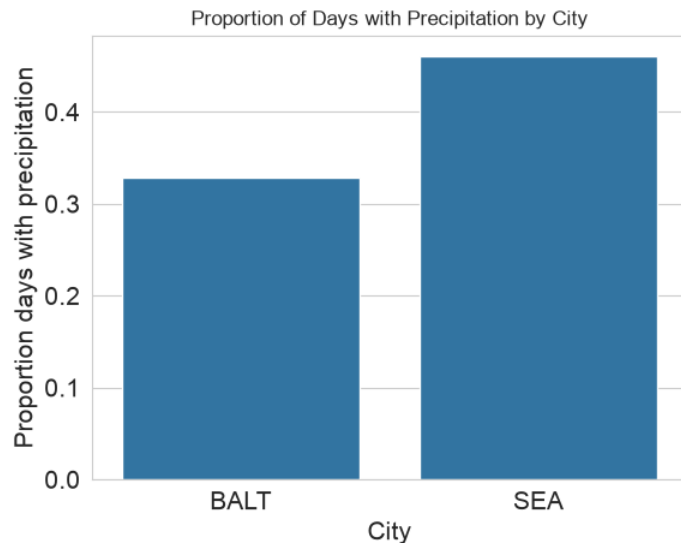


Figure 4: ‘Proportion of Days with Precipitation by City.’ Bar graph shows the proportion of days with precipitation of Baltimore (left) and Seattle (right).

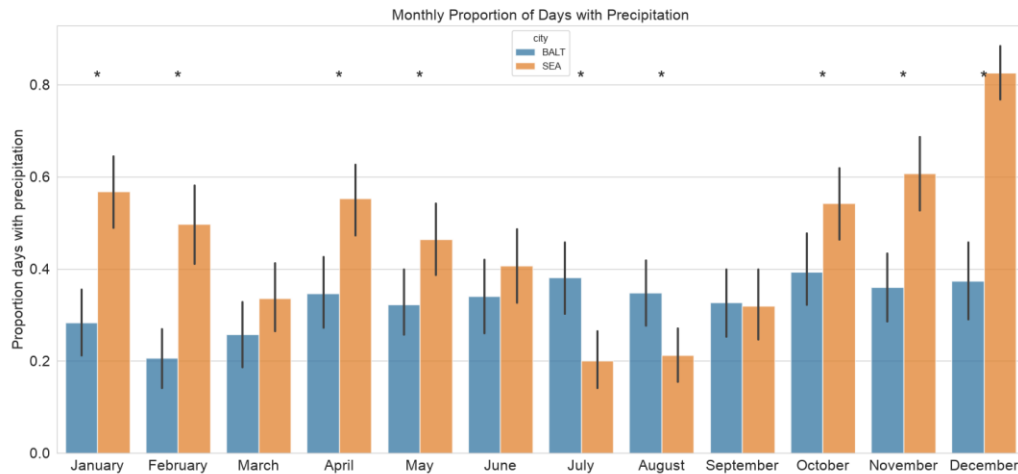


Figure 5: “Monthly Proportion of Days with Precipitation.” This graph plots the proportion of days with precipitation per month for each city, with Baltimore in blue and Seattle in orange. It includes confidence intervals and asterisks above months determined to have a statistically significant difference in proportion.